Large Language Model Driven Construction of a Finite–Volume Quantum Yang-Mills Theory

MathePhysics

MathePhysics is making public our LLM-driven efforts to resolve the four-dimensional Yang-Mills (YM) existence problem[1] (the first of many steps needed to establish a rigorous description of the Standard Model). We are constructing a YM QFT by pursuing renormalisation of Wilson Loops in 4D, continuing the programme Bałaban began in the 1980s.

In particular, we aim to compute the correlation functions of these loops. Note that this would not prove that the Wilson loops themselves constitute well-defined random variables, nor is our effort a proposed construction of the continuum YM measure, both of which are strictly stronger conditions than required by physical considerations (see research outline).

Our YM efforts began on 26/08/26 as part of a series of projects whose motivations are:

1. to understand the capacity of LLMs to push the frontiers and ambitions of academic inquiry while operating in the resource-constrained space academia usually occupies.

2. to examine and contrast the speed, accuracy, and cost of LLM-assisted science when guided by human expertise versus when left to raw and/or unguided methods.

3. to test the capacity of LLMs to overcome the methodological gaps that prevent meaningful communication even between experts in closely related fields.

Beyond our strong personal interest in the problem, we began with YM because it is ideally suited to these goals. It is fundamentally a product of the human desire to understand the universe, yet lacks even a formal consensus on what its precise statement should be. Further, the last best attempt (Bałaban's) is considered prohibitively dense to build upon by even the foremost experts – to the extent that many believed it required too steep a time investment to be a rational pursuit in the professional academic environment.

[1] Jaffe, Arthur, and Edward Witten. "Quantum Yang-Mills theory." Clay Mathematics Institute, 2000.

Finally, the commercial interest driving the unprecedented investment in solving problems with minuscule short-term "utility" (like YM) could cease anytime, pushing these problems back to the resource scarcity typical for them. The academic community rapidly needs to find the best ethos for using LLMs in the resource-constrained settings that are likely coming or the structural damage it suffers once the "interest bubble" bursts could be systemic.

In line with the research and AI/ML ethos of MathePhysics, we commit to complete and concurrent disclosure of all the tools, strategies, and methods used at every stage of this project. The work will be backed by human and LEAN verification (both ongoing). Given YM's significance, upon verification of our construction, we also commit to writing a manuscript at the highest pedagogical standards we are capable of. We will not (cannot!) claim personal credit for the technical achievements already made (or to be made) and limit our control and efforts to faithful stewardship of this project and its github repository.

To this end, our announcement is paired with two other documents: a human-generated outline for the general physics and mathematics public, which summarises the proof strategy we designed for LLMs; and an LLM-generated draft establishing the first RG step for Wilson loops, extracting the correct perimeter and corner renormalisations and a nonlocal error term. The estimates we prove upon future refinement should iterate across RG steps and allow us to prove convergence of the renormalised expectations in the UV limit for finite volume.

The repo also contains rough proof sketches comprised of previous work as well as provisional future constructions (modulo small errors) – these are not fully LLM-verified and have not been incorporated into our primary manuscript draft. Both academic and commercial communities are welcome to use the results we have made available to pursue better ones. Finally, we will be making public the entire chat history associated with this attempt.

Unquestionably, sans Claude and ChatGPT, this project's output would have been null. The human expertise was strictly limited to designing, implementing, and constantly refining the basic research outline – providing the "shape" of Mathematics, while the LLMs handled the work. All results are solely produced via retail subscriptions of Claude and ChatGPT.

We regret and apologise that our first scholarly work on this topic has had to be "AI slop" that only burdens the mathematical physics community further – in ordinary times,

we would have never dreamt of releasing a manuscript we could not defend each word of.

However, recent announcements on YM by people associated with OpenAI/Anthropic (which are far less resource-constrained than we are) have forced this release significantly ahead of schedule. We note that our release is seemingly the **only** set of publicly available documents that claims to have a path to resolving YM (OpenAI has not made its work public and other teams with concurrent claims have not revealed their specific work in 4D).

We have not made use of any progress made by concurrent attempts (Sec. 4, research outline). We have partially chosen this path so we could test whether expertise+LLM-driven research can compete with brute force LLM methods made possible by academically unviable sums of money. The total expenses incurred during this project are bounded above at $500. Even when including personnel salaries and pricing tokens at API equivalents, the total costs are of $\mathcal{O}$($10,000) or less in every calculation we are able to imagine.

Our only possible advantage lies in our methodology – we began with a human-driven research plan with clear ideas and outlines of what we were attempting to achieve, with the LLMs being used for feedback, refinement, and implementation; instead of brute-forcing our way to a solution. Thus, we profited from requiring fewer compute resources by virtue of being able to track and validate progress somewhat in real time – while carrying the risk of our efforts being completely in vain if our proof sketch is faulty and not well thought out.

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